

# Lecture 7 - Sep 16

## Pigeonhole Principle (Strong Form)

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If  $q_1 + q_2 + \dots + q_n = n + 1$  or more objects are placed into  $n$  boxes, then

box 1 has  $q_1$ , or more of the objects, or

box 2 has  $q_2$  or more of the objects, or

...

box  $n$  has  $q_n$  or more of the objects.

## Proof by contradiction

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Suppose box  $i$  contains at most  $q_i - 1$  of the objects.

For  $i = 1, \dots, n$ . Then we have at most  $(q_1 - 1) + (q_2 - 1) + \dots + (q_n - 1)$

$= q_1 + q_2 + \dots + q_n - n$  objects. In total, a contradiction.  $\square$

*Remarks:*

1. If  $m$  objects are placed into  $n$  boxes, then some box must contain at least  $\lceil \frac{m}{n} \rceil$  objects.

2. Given  $n$  integers  $a_1, \dots, a_n$ , If  $\frac{a_1 + a_2 + \dots + a_n}{n} < r$ , then  $\exists i$  such that  $a_i < r$ . Similarly, If  $\frac{a_1 + a_2 + \dots + a_n}{n} > r$ , then  $\exists i$  such that  $a_i > r$ . (and same for  $\geq r$ ).

Another way to look at it:

$$\begin{aligned} \frac{g * n}{n} &< r \\ \rightarrow g * n &< n * r \rightarrow g < r \end{aligned}$$

And  $g$  is the smallest number any  $a_i$  could possibly be.

## Example 1

Suppose there are  $n^2 + 1$  people in a straight line. Then it is always possible to choose  $n + 1$  of them one step forward so that going left to right, their heights are either increasing or decreasing.

## Theorem - Erdos-Szekeres 1935.

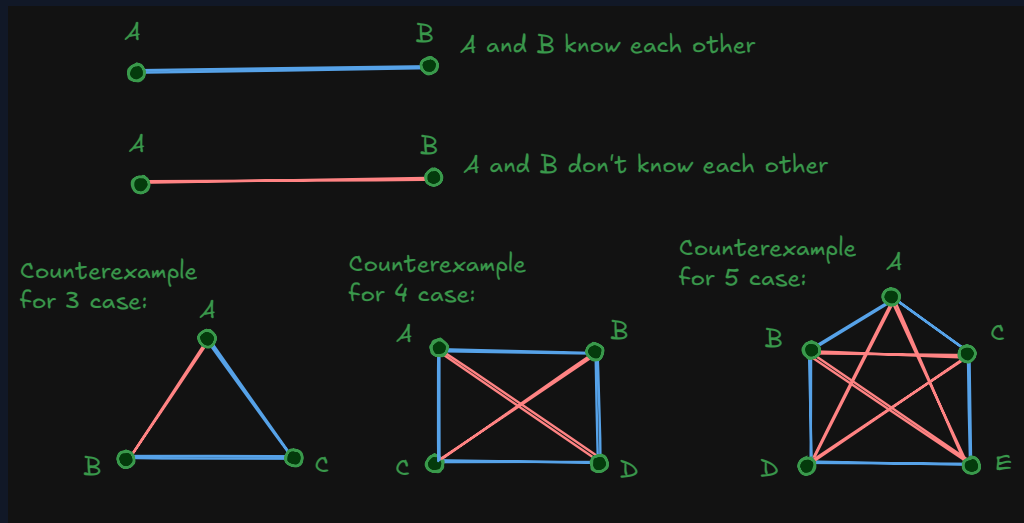
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Every sequence  $a_1, \dots, a_{n^2+1}$  of  $n^2 + 1$  real numbers, contains either an increasing or decreasing subsequence of length  $n + 1$ .

## Ramsey Theory - Applications of the strong PP

If six or more people, either there are three, each pair of whom are acquainted, or each pair of whom are unacquainted.

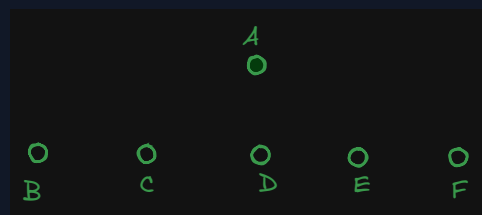
Why six?



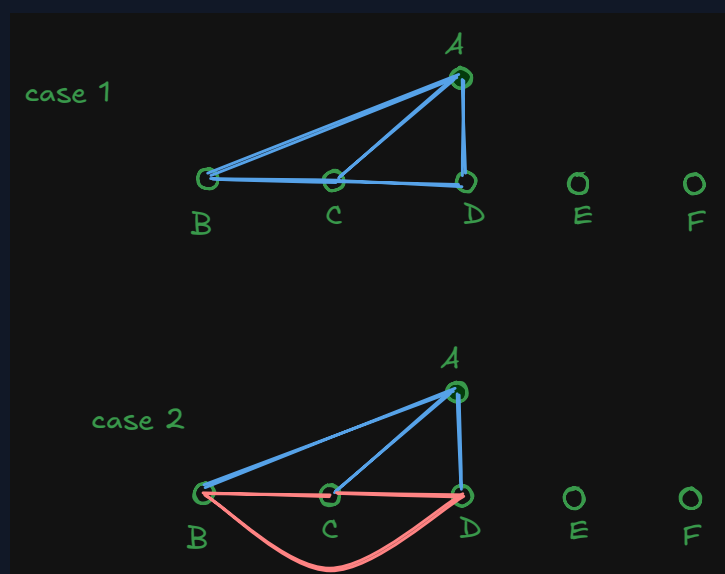
## Proof

Let  $A, B, C, D, E, F$  be six people.

Single A out. By the Strong PP, at least  $\lceil \frac{5}{2} \rceil = 3$  of  $B, C, D, E, F$  are either acquainted or unacquainted w/  $A$ .



By symmetry, we may assume that  $B, C, D$  know  $A$ :



Therefore, there are 2 cases, one where  $B, C, D$  or  $B, D$  know each other, in which case a blue triangle forms. In the latter case, none of  $B, C, D$  know each other, but in doing so form a red triangle.

Another phrasing: we may assume no two of  $B, C, D$  are acquainted. Then  $B, C, D$  are mutually unacquainted.  $\square$

## Graph theory notation

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$$k_n \rightarrow (k_s, k_t)$$

Among  $n$  or more people, there are  $s$  of them that are mutually acquainted or there are  $t$  of them that are mutually unacquainted.

### The ramsey number

$$R(s, t) = \min\{n \mid k_n \rightarrow k_s, k_t\}$$

Remark: if  $k_n \rightarrow (k_s, k_t)$ , then  $k_n \rightarrow (k_t, k_s)$

$$R(3, 3) = 6 \iff$$

$$(3, 3) \leq 6$$

$$R(3, 3) \geq 6$$

because  $k_5 \not\rightarrow (k_3, k_3)$

Other ramsey numbers:

$$R(2, 2) = 2$$

$$k_1 \not\rightarrow (k_2, k_2)$$

$$R(2, 3) = 3 \quad (k_2 \not\rightarrow (k_2, k_3))$$

$$\boxed{R(2, t) = R(t, 2) = t}$$

### proof

For any ramsey number of the form  $R(2, t)$  we cannot connect any people (otherwise the 2 condition gets met).

Therefore, the minimum number of people must be  $t$ , since if none are connected then  $t$  must be unacquainted.

## Ramsey Theorem

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$K(s, t)$  always exists.

Determining  $R(s, t)$  is notoriously difficult.

There is also a variant with multiple states called *Multicolor Ramsey Theory*  $R(s, t, r), R(n_1, n_2, \dots, n_k)$

## Graph theory

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A graph  $G = (V, E)$  is a pair, where  $V$  is a set, and  $E \subseteq 2$ -element subsets of  $V$ .

- Elements in  $V$  are *vertices* of  $G$
- Elements in  $E$  are *edges* of  $G$

- $G$  is *finite* if  $|V|$  is finite
- Two vertices  $x, y \in V$  are adjacent in  $G$  if  $\{x, y\} \in E$ , but more used notation is  $xy \in E$  or  $yx \in E$
- The usual way to picture a graph use dots to represent vertices and lines to represent edges. Just how dots and lines are drawn are irrelevant.

$$G = (V, E), V = a, b, c, d, e$$

$$E = \{\{a, b\}, \{b, c\}, \{c, d\}, \{a, e\}\}$$

For each  $v \in V$ ,  $N(v) := \{u \in V, \{u, v\} \in E\}$

Called neighborhood of  $v \rightarrow$  degree of  $v = d(v) = |N(v)|$

$K_n$ : the complete graph on  $n$  vertices